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Wound-rotor electric machine

Abstract

A system for converting energy includes a doubly-fed induction machine including a rotor having a plurality of rotor windings, and a stator having a plurality of stator windings. The system also includes a power conversion unit in electrical communication with the plurality of rotor windings, where the power conversion unit is configured to excite the plurality of rotor windings with an alternating current. The system also includes a relay unit in electrical communication with the plurality of stator windings, where the relay unit is configured to electrically interconnect a plurality of terminals of the plurality of stator windings. The system also includes a controller in electrical communication with the power conversion unit and the relay unit, where the controller is programmed to adjust an operation of the doubly-fed induction machine using the power conversion unit and the relay unit based on the rotational velocity of the rotor.

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Background/Summary

INTRODUCTION
(1) The present disclosure relates to a system for converting energy, more specifically, to a narrowband independent-speed variable-frequency electric machine.
(2) To convert energy from mechanical to electrical or electrical to mechanical, electric machines (e.g., generators and motors) have been developed. Often, output power characteristics of electric machines are related to characteristics of the mechanical input. For example, in some electric machines, the frequency of the output power may be proportional to the rotational velocity of the electric machine. Often, specific characteristics of output power are desirable regardless of the characteristics of the mechanical input. For example, in certain applications, it may be desirable to provide output power within a frequency range and with a specific amplitude. However, systems which have been developed to regulate characteristics of output power of electric machines may be large, heavy, and prone to mechanical wear.
(3) Thus, while current systems for converting energy achieve their intended purpose, there is a need for a new and improved system for converting energy.
SUMMARY
(4) According to several aspects, a system for converting energy is provided. The system includes a doubly-fed induction machine including a rotor having a plurality of rotor windings, and a stator having a plurality of stator windings. The system also includes a power conversion unit in electrical communication with the plurality of rotor windings, where the power conversion unit is configured to excite the plurality of rotor windings with an alternating current. The system also includes a relay unit in electrical communication with the plurality of stator windings, where the relay unit is configured to electrically interconnect a plurality of terminals of the plurality of stator windings. The system also includes a controller in electrical communication with the power conversion unit and the relay unit, where the controller is programmed to determine a rotational velocity of the rotor and adjust an operation of the doubly-fed induction machine using the power conversion unit and the relay unit based on the rotational velocity of the rotor.
(5) In another aspect of the present disclosure, the plurality of rotor windings further includes a first three-phase rotor winding having a first rotor winding coil, a second rotor winding coil, and a third rotor winding coil. The plurality of rotor windings further includes a second three-phase rotor winding having a fourth rotor winding coil, a fifth rotor winding coil, and a sixth rotor winding coil. The plurality of stator windings further includes a first three-phase stator winding having a first stator winding coil having a first and second terminal, a second stator winding coil having a first and second terminal, and a third stator winding coil having a first and second terminal. The plurality of stator windings further includes a second three-phase stator winding having a fourth stator winding coil having a first and second terminal, a fifth stator winding coil having a first and second terminal, and a sixth stator winding coil having a first and second terminal. The system further includes a first, second, and third output terminal.

(6) In another aspect of the present disclosure, the power conversion unit further may include a first inverter having three outputs, where each output of the first inverter is in electrical communication with one of the first, second, or third rotor winding coils. The power conversion unit further may include a second inverter having three outputs, where each output of the second inverter is in electrical communication with one of the fourth, fifth, or sixth rotor winding coils. The first inverter and the second inverter are configured to be powered by a direct current (DC) power source.

(7) In another aspect of the present disclosure, the system further includes a slip ring configured to provide electrical communication between each output of the first inverter and one of the first, second, or third rotor winding coils and between each output of the second inverter and one of the fourth, fifth, or sixth rotor winding coils.

(8) In another aspect of the present disclosure, the relay unit further may include a plurality of solid-state relays in electrical communication with the first three-phase stator winding and the second three-phase stator winding.

(9) In another aspect of the present disclosure, to adjust the operation of the doubly-fed induction machine, the controller is further programmed to adjust an excitation frequency of the first three-phase rotor winding using the first inverter and the excitation frequency of the second three-phase rotor winding using the second inverter of the power conversion unit based on the rotational velocity of the rotor to regulate an output voltage frequency when the doubly-fed induction machine is used as a generator.

(10) In another aspect of the present disclosure, to adjust the operation of the doubly-fed induction machine, the controller is further programmed to configure the doubly-fed induction machine to have four effective poles using the power conversion unit and the relay unit based on the rotational velocity of the rotor to regulate an output voltage frequency when the doubly-fed induction machine is used as a generator. To adjust the operation of the doubly-fed induction machine, the controller is further programmed to configure the doubly-fed induction machine to have eight effective poles using the power conversion unit and the relay unit based on the rotational velocity of the rotor to regulate an output voltage frequency when the doubly-fed induction machine is used as a generator.

(11) In another aspect of the present disclosure, to configure the doubly-fed induction machine to have four effective poles, the controller is further programmed to configure the first inverter to excite the first rotor winding coil with a first sinusoidal current, excite the second rotor winding coil with a second sinusoidal current, where the second sinusoidal current lags the first sinusoidal current by 240 degrees, and excite the third rotor winding coil with a third sinusoidal current, where the third sinusoidal current lags the first sinusoidal current by 120 degrees. To configure the doubly-fed induction machine to have four effective poles, the controller is further programmed to configure the second inverter to excite the fourth rotor winding coil with a fourth sinusoidal current, where the fourth sinusoidal current is 180 degrees out of phase with the first sinusoidal current, excite the fifth rotor winding coil with a fifth sinusoidal current, where the fifth sinusoidal current is 180 degrees out of phase with the second sinusoidal current, and excite the sixth rotor winding coil with a sixth sinusoidal current, where the sixth sinusoidal current is 180 degrees out of phase with the third sinusoidal current. To configure the doubly-fed induction machine to have four effective poles, the controller is further programmed to configure the relay unit to electrically connect the first terminal of the first stator winding coil with the second output terminal and the first terminal of the sixth stator winding coil. To configure the doubly-fed induction machine to have four effective poles, the controller is further programmed to configure the relay unit to electrically connect the first terminal of the second stator winding coil with the third output terminal and the first terminal of the fifth stator winding coil. To configure the doubly-fed induction machine to have four effective poles, the controller is further programmed to configure the relay unit to electrically connect the first terminal of the third stator winding coil with the first

output terminal and the first terminal of the fourth stator winding coil. To configure the doubly-fed induction machine to have four effective poles, the controller is further programmed to configure the relay unit to electrically connect the second terminal of the fourth stator winding coil to the second terminal of the fifth stator winding coil and the second terminal of the sixth stator winding coil.

(12) In another aspect of the present disclosure, to configure the doubly-fed induction machine to have eight effective poles, the controller is further programmed to configure the first inverter to excite the first rotor winding coil with a first sinusoidal current, excite the second rotor winding coil with a second sinusoidal current, where the second sinusoidal current lags the first sinusoidal current by 120 degrees, and excite the third rotor winding coil with a third sinusoidal current, where the third sinusoidal current lags the second sinusoidal current by 120 degrees. To configure the doubly-fed induction machine to have eight effective poles, the controller is further programmed to configure the second inverter to excite the fourth rotor winding coil with the first sinusoidal current, excite the fifth rotor winding coil with the second sinusoidal current, and excite the sixth rotor winding coil with the third sinusoidal current. To configure the doubly-fed induction machine to have eight effective poles, the controller is further programmed to configure the relay unit to electrically connect the first terminal of the first stator winding coil with the third output terminal and the second terminal of the sixth stator winding coil. To configure the doubly-fed induction machine to have eight effective poles, the controller is further programmed to configure the relay unit to electrically connect the first terminal of the second stator winding coil with the second output terminal and the second terminal of the fifth stator winding coil. To configure the doubly-fed induction machine to have eight effective poles, the controller is further programmed to configure the relay unit to electrically connect the first terminal of the third stator winding coil with the first output terminal and the second terminal of the fourth stator winding coil. To configure the doubly-fed induction machine to have eight effective poles, the controller is further programmed to configure the relay unit to electrically connect the first terminal of the fourth stator winding coil to the first terminal of the fifth stator winding coil and the first terminal of the sixth stator winding coil.

(13) According to several aspects, a system for converting energy is provided. The system includes a doubly-fed induction machine including a rotor having a plurality of rotor windings, and a stator having a plurality of stator windings. The system also includes a power conversion unit in wireless electrical communication with the plurality of rotor windings, where the power conversion unit is configured to excite the plurality of rotor windings with an alternating current. The system also includes a relay unit in electrical communication with the plurality of stator windings, where the relay unit is configured to electrically interconnect a plurality of terminals of the plurality of stator windings. The system also includes a controller in electrical communication with the power conversion unit and the relay unit, where the controller is programmed to determine a rotational velocity of the rotor and adjust an operation of the doubly-fed induction machine using the power conversion unit and the relay unit based on the rotational velocity of the rotor.

(14) In another aspect of the present disclosure, the plurality of rotor windings further includes a first three-phase rotor winding having a first rotor winding coil, a second rotor winding coil, and a third rotor winding coil. The plurality of rotor windings further includes a second three-phase rotor winding having a fourth rotor winding coil, a fifth rotor winding coil, and a sixth rotor winding coil. The plurality of stator windings further includes a first three-phase stator winding having a first stator winding coil having a first and second terminal, a second stator winding coil having a first and second terminal, and a third stator winding coil having a first and second terminal. The plurality of stator windings further includes a second three-phase stator winding having a fourth stator winding coil having a first and second terminal, a fifth stator winding coil having a first and second terminal, and a sixth stator winding coil having a first and second terminal. The system further includes a first, second, and third output terminal.

(15) In another aspect of the present disclosure, the power conversion unit further may include a first inverter having three outputs, where each output of the first inverter is in electrical communication with one of the first, second, or third rotor winding coils. The power conversion unit further may include a second inverter having three outputs, where each output of the second inverter is in electrical communication with one of the fourth, fifth, or sixth rotor winding coils. The first inverter and the second inverter are affixed to the rotor.
(16) In another aspect of the present disclosure, the power conversion unit further may include a wireless power transfer system including a stationary power transfer inverter and a stationary wireless power transfer coil in electrical communication with the stationary power transfer inverter. The wireless power transfer system further includes a rotating power transfer rectifier affixed to the rotor and a rotating wireless power transfer coil affixed to the rotor, where the rotating wireless power transfer coil is in electrical communication with the rotating power transfer rectifier and the first inverter and the second inverter of the power conversion unit. The stationary wireless power transfer coil is inductively coupled to the rotating wireless power transfer coil to transmit power between the stationary power transfer inverter and the rotating power transfer rectifier. The power conversion unit further may include a wireless data transfer system including a stationary wireless data transceiver and a rotating wireless data transceiver affixed to the rotor. The wireless data transfer system further includes a rotating controller affixed to the rotor, where the rotating controller is in electrical communication with the rotating wireless data transceiver and the first inverter and the second inverter of the power conversion unit. The rotating controller is programmed to receive command data from the stationary wireless data transceiver using the rotating wireless data transceiver and adjust an operation of the first inverter and the second inverter of the power conversion unit based at least in part on the command data.
(17) In another aspect of the present disclosure, the relay unit further may include a plurality of solid-state relays in electrical communication with the first three-phase stator winding and the second three-phase stator winding.
(18) In another aspect of the present disclosure, to adjust the operation of the doubly-fed induction machine, the controller is further programmed to adjust an excitation frequency of the first three-phase rotor winding using the first inverter and the excitation frequency of the second three-phase rotor winding using the second inverter of the power conversion unit based on the rotational velocity of the rotor to regulate an output voltage frequency when the doubly-fed induction machine is used as a generator.
(19) In another aspect of the present disclosure, to adjust the operation of the doubly-fed induction machine, the controller is further programmed to configure the doubly-fed induction machine to have four effective poles using the power conversion unit and the relay unit based on the rotational velocity of the rotor to regulate an output voltage frequency when the doubly-fed induction machine is used as a generator. To adjust the operation of the doubly-fed induction machine, the controller is further programmed to configure the doubly-fed induction machine to have eight effective poles using the power conversion unit and the relay unit based on the rotational velocity of the rotor to regulate an output voltage frequency when the doubly-fed induction machine is used as a generator.
(20) In another aspect of the present disclosure, to configure the doubly-fed induction machine to have four effective poles, the controller is further programmed to configure the first inverter to excite the first rotor winding coil with a first sinusoidal current, excite the second rotor winding coil with a second sinusoidal current, where the second sinusoidal current lags the first sinusoidal current by 240 degrees, and excite the third rotor winding coil with a third sinusoidal current, where the third sinusoidal current lags the first sinusoidal current by 120 degrees. To configure the doubly-fed induction machine to have four effective poles, the controller is further programmed to configure the second inverter to excite the fourth rotor winding coil with a fourth sinusoidal current, where the fourth sinusoidal current is 180 degrees out of phase with the first sinusoidal

current, excite the fifth rotor winding coil with a fifth sinusoidal current, where the fifth sinusoidal current is 180 degrees out of phase with the second sinusoidal current, and excite the sixth rotor winding coil with a sixth sinusoidal current, where the sixth sinusoidal current is 180 degrees out of phase with the third sinusoidal current. To configure the doubly-fed induction machine to have four effective poles, the controller is further programmed to configure the relay unit to electrically connect the first terminal of the first stator winding coil with the second output terminal and the first terminal of the sixth stator winding coil. To configure the doubly-fed induction machine to have four effective poles, the controller is further programmed to configure the relay unit to electrically connect the first terminal of the second stator winding coil with the third output terminal and the first terminal of the fifth stator winding coil. To configure the doubly-fed induction machine to have four effective poles, the controller is further programmed to configure the relay unit to electrically connect the first terminal of the third stator winding coil with the first output terminal and the first terminal of the fourth stator winding coil. To configure the doubly-fed induction machine to have four effective poles, the controller is further programmed to configure the relay unit to electrically connect the second terminal of the fourth stator winding coil to the second terminal of the fifth stator winding coil and the second terminal of the sixth stator winding coil.

(21) In another aspect of the present disclosure, to configure the doubly-fed induction machine to have eight effective poles, the controller is further programmed to configure the first inverter to excite the first rotor winding coil with a first sinusoidal current, excite the second rotor winding coil with a second sinusoidal current, where the second sinusoidal current lags the first sinusoidal current by 120 degrees, and excite the third rotor winding coil with a third sinusoidal current, where the third sinusoidal current lags the second sinusoidal current by 120 degrees. To configure the doubly-fed induction machine to have eight effective poles, the controller is further programmed to configure the second inverter to excite the fourth rotor winding coil with the first sinusoidal current, excite the fifth rotor winding coil with the second sinusoidal current, and excite the sixth rotor winding coil with the third sinusoidal current. To configure the doubly-fed induction machine to have eight effective poles, the controller is further programmed to configure the relay unit to electrically connect the first terminal of the first stator winding coil with the third output terminal and the second terminal of the sixth stator winding coil. To configure the doubly-fed induction machine to have eight effective poles, the controller is further programmed to configure the relay unit to electrically connect the first terminal of the second stator winding coil with the second output terminal and the second terminal of the fifth stator winding coil. To configure the doubly-fed induction machine to have eight effective poles, the controller is further programmed to configure the relay unit to electrically connect the first terminal of the third stator winding coil with the first output terminal and the second terminal of the fourth stator winding coil. To configure the doubly-fed induction machine to have eight effective poles, the controller is further programmed to configure the relay unit to electrically connect the first terminal of the fourth stator winding coil to the first terminal of the fifth stator winding coil and the first terminal of the sixth stator winding coil.

(22) According to several aspects, an electric machine is provided. The electric machine includes a stator having a hollow cylindrical shape, the stator including a plurality of stator slots disposed at a uniform pitch along a circumference of the stator. The stator further includes a first, second, third, fourth, fifth, and sixth stator coil, where each stator coil has a first stator coil loop electrically connected in series to a second stator coil loop. For each of the first, second, third, fourth, fifth, and sixth stator coil, the first stator coil loop is disposed in a first pair of stator slots, where a second pair of stator slots is disposed between the first pair of stator slots. The second stator coil loop is disposed in a third pair of stator slots, where a fourth pair of stator slots is disposed between the third pair of stator slots, and where the second stator coil loop is diametrically opposed to the first stator coil loop. The electric machine also includes a rotor disposed rotatably inside the stator along

a central axis of the stator, the rotor including a plurality of rotor slots disposed at a uniform pitch along a circumference of the rotor. The rotor further includes a first, second, third, fourth, fifth, and sixth rotor coil. Each rotor coil has a first rotor coil loop electrically connected in series to a second rotor coil loop. For each of the first, second, third, fourth, fifth, and sixth rotor coil, the first rotor coil loop is disposed in a first pair of rotor slots, where a second pair of rotor slots is disposed between the first pair of rotor slots. The second rotor coil loop is disposed in a third pair of rotor slots, where a fourth pair of rotor slots is disposed between the third pair of rotor slots, and where the second rotor coil loop is diametrically opposed to the first rotor coil loop.
(23) In another aspect of the present disclosure, The electric machine further includes a controller affixed to an exterior surface of the stator, where the controller is programmed to determine a rotational velocity of the rotor and electrically change a pole number of the rotor and the stator based at least in part on the rotational velocity of the rotor by adjusting a phase angle of each phase of a three-phase current supply electrically connected to the rotor coils and changing a plurality of electrical connections between each of the stator coils.
(24) Further areas of applicability will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present disclosure in any way.
(2) FIG. **1** is a block diagram of a system for converting energy according to an exemplary embodiment;
(3) FIG. **2**A is a diagram of a winding pattern for a stator of an electric machine according to an exemplary embodiment;
(4) FIG. **2**B is a diagram of a winding pattern for a rotor of an electric machine according to an exemplary embodiment;
(5) FIG. **3** is a schematic diagram of a relay unit connected to stator windings of an electric machine according to an exemplary embodiment;
(6) FIG. **4** is a block diagram of an alternate system for converting energy according to an exemplary embodiment; and
(7) FIG. **5** is a graph of a mode of operation of a system for converting energy according to an exemplary embodiment;
DETAILED DESCRIPTION
(8) The following description is merely exemplary in nature and is not intended to limit the present disclosure, application, or uses.
(9) Referring to FIG. **1**, a block diagram of a system for converting energy is illustrated and generally indicated by reference numeral **10**. The system **10** generally includes a doubly-fed induction machine **12**, a power conversion unit **14**, a plurality of slip rings **15**, a relay unit **16**, and a controller **18**.
(10) The doubly-fed induction machine **12** is used to convert mechanical energy to electrical energy, and/or electrical energy to mechanical energy. The doubly-fed induction machine **12** includes a stator **20** and a rotor **22**. The stator **20** has a hollow cylindrical shape, and the rotor **22** has a solid cylindrical shape. The rotor **22** is suspended rotatably along a central axis of the stator **20**. The structure of the stator **20** and the rotor **22** will be discussed in further detail in reference to FIG. **2**. In a non-limiting example, when the doubly-fed induction machine **12** is used to convert mechanical energy to electrical energy (i.e., as a generator), the rotor **22** is affixed to an source of

rotational mechanical energy, such as an internal combustion engine, a jet engine, a wind turbine or the like. In another non-limiting example, when the doubly-fed induction machine **12** is used to convert electrical energy to mechanical energy, stator **20** is electrically connected to a source of electrical energy to rotate the rotor **22** and the rotor **22** is affixed to a mechanical load, and the rotor **22** rotates to provide energy to the mechanical load. In the present disclosure, the term "doubly-fed" means that both the stator **20** and the rotor **22** of the doubly-fed induction machine **12** may supplied with electrical energy.

(11) The stator **20** is a stationary component of the doubly-fed induction machine **12** which converts electrical energy to magnetic field energy and/or magnetic field energy to electrical energy. The stator **20** includes a first three-phase stator winding **24** and a second three-phase stator winding **26**. The first three-phase stator winding **24** has a first stator winding coil **28**, a second stator winding coil **30**, and a third stator winding coil **32**. The second three-phase stator winding **26** has a fourth stator winding coil **34**, a fifth stator winding coil **36**, and a sixth stator winding coil **38**. The first three-phase stator winding **24** and the second three-phase stator winding **26** are in electrical communication with the relay unit **16**. The electrical connections of the stator **20** will be discussed in more detail in reference to FIGS. **2** and **3**. In a non-limiting example, when the three-phase stator windings **24**, **26** are in the presence of a changing magnetic field, electric currents are induced in the three-phase stator windings **24**, **26**, converting the energy of the changing magnetic field to electrical energy. In another non-limiting example, when the three-phase stator windings **24**, **26** are excited by a changing electric field, a changing magnetic field is induced near the three-phase stator windings **24**, **26**, converting the energy of the changing electric field to a changing magnetic field.

(12) The rotor **22** is a rotating component of the doubly-fed induction machine **12** which converts electrical energy to magnetic field energy and/or magnetic field energy to electrical energy. The rotor **22** includes a first three-phase rotor winding **40** and a second three-phase rotor winding **42**. The first three-phase rotor winding **40** has a first rotor winding coil **44**, a second rotor winding coil **46**, and a third rotor winding coil **48**. The second three-phase rotor winding **42** has a fourth rotor winding coil **50**, a fifth rotor winding coil **52**, and a sixth rotor winding coil **54**. The first three-phase rotor winding **40** has a first, second, and third winding terminal **56**, **58**, **60** which are used to connect the first three-phase rotor winding **40** to the power conversion unit **14** using the plurality of slip rings **15**. The second three-phase rotor winding **42** has a fourth, fifth, and sixth winding terminal **62**, **64**, **66** which are used to connect the second three-phase rotor winding **42** to the power conversion unit **14** using the plurality of slip rings **15**.

(13) In the exemplary embodiment shown in FIG. **1**, the first, second, and third rotor winding coils **44**, **46**, **48** are connected in a delta configuration. In the delta configuration, a first terminal of the first rotor winding coil **44** is connected to a first terminal of the second rotor winding coil **46**, a second terminal of the second rotor winding coil **46** is connected to a first terminal of the third rotor winding coil **48**, and a second terminal of the third rotor winding coil **48** is connected to a second terminal of the first rotor winding coil **44**. The first winding terminal **56** is connected to the first terminals of the first rotor winding coil **44** and the second rotor winding coil **46**. The second winding terminal **58** is connected to the second terminal of the second rotor winding coil **46** and the first terminal of the third rotor winding coil **48**. The third winding terminal **60** is connected to the second terminal of the third rotor winding coil and the second terminal of the first rotor winding coil **44**. The fourth, fifth, and sixth rotor winding coils **50**, **52**, **54** are connected to the fourth, fifth, and sixth winding terminals **62**, **64**, **66** in the same manner as described above for the first, second, and third rotor winding coils **44**, **46**, **48**. In other exemplary embodiments, the rotor winding coils **44**, **46**, **48**, **50**, **52**, **54** are connected in two wye configurations.

(14) In a non-limiting example, when the three-phase rotor windings **40**, **42** are in the presence of a changing magnetic field, electric currents are induced in the three-phase rotor windings **40**, **42**, converting the energy of the changing magnetic field to electrical energy. In another non-limiting

example, when the three-phase rotor windings **40**, **42** are excited by a changing electric field, a changing magnetic field is induced near the three-phase rotor windings **40**, **42**, converting the energy of the changing electric field to a changing magnetic field.

(15) The power conversion unit **14** is used to convert a direct current (DC) power supply **72***a*, **72***b* to two three-phase alternating current (AC) power sources and provide a three-phase AC power source to each of the three-phase rotor windings **40**, **42**. In the exemplary embodiment shown in FIG. **1**, the power conversion unit **14** includes a first inverter **68** and a second inverter **70**. The inverters **68**, **70** convert DC power to AC power and allow control of various characteristics of the AC power by the controller **18**. In a non-limiting example, the inverters **68**, **70** respond to data signals from the controller **18** by adjusting an amplitude, frequency, and/or phase angle of the AC current and/or voltage output. The inverters **68**, **70** are connected to the DC power supply **72***a*, **72***b* and provide an AC power to each of the rotor winding terminals **56**, **58**, **60**, **62**, **64**, **66** using the plurality of slip rings **15**. The inverters **68**, **70** are also in electrical communication with the controller **18**. It should be understood that additional methods of communication between the inverters **68**, **70** and the controller **18**, including wireless, wired, electrical, optical, and/or opto-electrical communication are within the scope of the present disclosure. In an exemplary embodiment, the inverters **68**, **70** are square wave, modified sine wave, pulsed sine wave, pulse width modulated (PWM) wave or pure sine wave inverters. It should be understood that additional types of inverters or other electrical devices designed to convert DC power to AC power are within the scope of the present disclosure.

(16) The plurality of slip rings **15** are used to transmit AC power between the power conversion unit **14** and the rotor **22** of the doubly-fed induction machine **12**. The slip rings **15** are electromechanical devices allowing transmission of power from a stationary structure to a rotating structure. In an exemplary embodiment, the slip rings include six conductive rings affixed to the rotor **22** and six conductive contacts affixed to the power conversion unit **14**. The conductive contacts make contact with the conductive rings to form six continuous electrical conduction paths between the power conversion unit **14** and the rotor winding terminals **56**, **58**, **60**, **62**, **64**, **66**.

(17) The relay unit **16** is used to switch the electrical connections between the stator winding coils **28**, **30**, **32**, **34**, **36**, **38** and a first relay unit output **74**, a second relay unit output **76**, and a third relay unit output **78**. An exemplary embodiment of the relay unit **16** and the connections of the relay unit **16** to the stator winding coils **28**, **30**, **32**, **34**, **36**, **38** will be discussed in greater detail in reference to FIG. **3**. The relay unit **16** is in electrical communication with the stator winding coils **28**, **30**, **32**, **34**, **36**, **38**, the controller **18**, and the relay unit outputs **74**, **76**, **78**.

(18) The controller **18** is used to control the operation of the doubly-fed induction machine **12** using the power conversion unit **14** and the relay unit **16**. The controller **18** includes at least one processor **80** and a non-transitory computer readable storage device or media **82**. The processor **80** may be a custom made or commercially available processor, a central processing unit (CPU), a graphics processing unit (GPU), an auxiliary processor among several processors associated with the controller **18**, a semiconductor-based microprocessor (in the form of a microchip or chip set), a macroprocessor, a combination thereof, or generally a device for executing instructions. The computer readable storage device or media **82** may include volatile and nonvolatile storage in read-only memory (ROM), random-access memory (RAM), and keep-alive memory (KAM), for example. KAM is a persistent or non-volatile memory that may be used to store various operating variables while the processor **80** is powered down. The computer-readable storage device or media **82** may be implemented using a number of memory devices such as PROMs (programmable read-only memory), EPROMs (electrically PROM), EEPROMs (electrically erasable PROM), flash memory, or another electric, magnetic, optical, or combination memory devices capable of storing data, some of which represent executable instructions, used by the controller **18** to control the system **10**. The controller **18** may also consist of multiple controllers which are in electrical communication with each other.

(19) The controller **18** is in electrical communication with the power conversion unit **14** and the relay unit **16**. The media **82** of the controller **18** contains executable instructions to control the operation of the doubly-fed induction machine **12** using the power conversion unit **14** and the relay unit **16**.

(20) Referring to FIG. **2**A, a diagram of a winding pattern for the stator **20** according to an exemplary embodiment is illustrated and generally indicated by reference numeral **90**. In the exemplary embodiment shown in FIG. **2**A, the stator **20** is a hollow cylindrical shape having a plurality of stator slots **92** disposed at a uniform pitch along a circumference of the stator **20**. In the exemplary embodiment of FIG. **2**A, the stator **20** has 24 stator slots **92**. It should be understood that alternative embodiments including a different number of stator slots **92** are included within the scope of the present disclosure. The stator winding pattern **90** contains the six stator winding coils **28**, **30**, **32**, **34**, **36**, **38**, each having a first stator coil loop **94** and a second stator coil loop **96** connected in series. It should be understood that alternative embodiments including more than two stator coil loops and/or stator coil loops connected in series and/or parallel are within the scope of the present disclosure. The first and second stator coil loops **94**, **96** include a plurality of loops of wire disposed in a pair of the plurality of stator slots **92**. For example, the first stator coil loop **94** is disposed in a first stator slot **98** and a second stator slot **100**. The first stator slot **98** is separated from the second stator slot **100** by a third stator slot **102** and a fourth stator slot **104**. The second stator coil loop **96** is disposed in a fifth stator slot **106** and a sixth stator slot **108**. The fifth stator slot **106** is separated from the sixth stator slot **108** by a seventh stator slot **110** and an eighth stator slot **112**. Accordingly, the first stator coil loop **94** is diametrically opposed to the second stator coil loop **96**. The aforementioned pattern is repeated for each of the stator coil loops of the six stator winding coils **28**, **30**, **32**, **34**, **36**, **38** as shown in FIG. **2**A. The stator **20** has a central axis **114** located at the center of the hollow portion of the stator **20**.

(21) Referring to FIG. **2**B, a diagram of a winding pattern for the rotor **22** according to an exemplary embodiment is illustrated and generally indicated by reference numeral **120**. In the exemplary embodiment shown in FIG. **2**B, the rotor **22** is a solid cylindrical shape having a plurality of rotor slots **122** disposed at a uniform pitch along a circumference of the rotor **22**. The rotor winding pattern **120** contains six rotor winding coils **44**, **46**, **48**, **50**, **52**, **54**, each having a first rotor coil loop **124** and a second rotor coil loop **126**. The first and second rotor coil loops **124**, **126** are connected in series. It should be understood that alternative embodiments including more than two rotor coil loops and/or rotor coil loops connected in series and/or parallel are within the scope of the present disclosure. The first and second rotor coil loops **124**, **126** include a plurality of loops of wire disposed in a pair of the plurality of rotor slots **122**. For example, the first rotor coil loop **124** is disposed in a first rotor slot **128** and a second rotor slot **136**. The first rotor slot **128** is separated from the second rotor slot **136** by a third rotor slot **132** and a fourth rotor slot **134**. The second rotor coil loop **126** is disposed in a fifth rotor slot **136** and a sixth rotor slot **138**. The fifth rotor slot **136** is separated from the sixth rotor slot **138** by a seventh rotor slot **140** and an eighth rotor slot **142**. The aforementioned pattern is repeated for each of the rotor coil loops of the six rotor winding coils **44**, **46**, **48**, **50**, **52**, **54**. The rotor **22** has a rotor shaft **144** located along a central axis of the rotor **22**. The rotor shaft **144** is used to rotatably secure the rotor **22** along the central axis **114** of the stator **20**. The rotor shaft **144** is also used to mechanically affix the rotor **22** to a source of mechanical energy or a mechanical load, as discussed above.

(22) Referring to FIG. **3**, a schematic diagram of the relay unit **16** connected to the three-phase stator windings **24**, **26** according to an exemplary embodiment is illustrated and generally indicated by reference numeral **150**. In the exemplary embodiment of FIG. **3**, the relay unit **16** contains six single pole, double throw (SPDT) relays **156**, **160**, **164**, **166**, **170**, **178** and two single pole, single throw (SPST) relays **172**, **176**. It should be understood that alternative embodiments of the relay unit **16** including alternative configurations, types, and/or number of relays are within the scope of the present disclosure. Furthermore, realizations of the relay unit **16** using electromagnetic relays,

solid-state relays, and/or additional types of electronically controllable relays are within the scope of the present disclosure.

(23) The first stator winding coil **28** has a first terminal **152***a* and a second terminal **152***b*. The first terminal **152***a* is connected to the second stator winding coil **30**, and the second terminal **152***b* is connected to the first relay unit output **74**. The second stator winding coil **30** has a first terminal **154***a* and a second terminal **154***b*. The first terminal **154***a* is connected to the first terminal **152***a* of the first stator winding coil **28**, and the second terminal **154***b* is connected to a first relay **156**. The third stator winding coil **32** has a first terminal **158***a* and a second terminal **158***b*. The first terminal **158***a* is connected to the first terminal **154***a* of the second stator winding coil **30**, and the second terminal **158***b* is connected to a second relay **160**.

(24) The first relay **156** has a pole terminal **156***a*, a first switched terminal **156***b*, and a second switched terminal **156***c*. The pole terminal **156***a* is connected to the second terminal **154***b* of the second stator winding coil **30**. The first switched terminal **156***b* is connected to the second relay unit output **76**. The second switched terminal **156***c* is connected to the third relay unit output **78**.

(25) The second relay **160** has a pole terminal **160***a*, a first switched terminal **160***b*, and a second switched terminal **160***c*. The pole terminal **160***a* is connected to the second terminal **158***b* of the third stator winding coil **32**. The first switched terminal **160***b* is connected to the third relay unit output **78**. The second switched terminal **160***c* is connected to the second relay unit output **76**.

(26) The fourth stator winding coil **34** has a first terminal **162***a* and a second terminal **162***b*. The first terminal **162***a* is connected to a third relay **164**, and the second terminal **162***b* is connected to a fourth relay **166**. The fifth stator winding coil **36** has a first terminal **168***a* and a second terminal **168***b*. The first terminal **168***a* is connected to a fifth relay **170**, and the second terminal **168***b* is connected to a sixth relay **172**. The sixth stator winding coil **38** has a first terminal **174***a* and a second terminal **174***b*. The first terminal **174***a* is connected to a seventh relay **176**, and the second terminal **174***b* is connected to an eighth relay **178**.

(27) The third relay **164** has a pole terminal **164***a*, a first switched terminal **164***b*, and a second switched terminal **164***c*. The pole terminal **164***a* is connected to the first terminal **162***a* of the fourth stator winding coil **34**. The first switched terminal **164***b* is connected to first terminal **168***a* of the fifth stator winding coil **36**. The second switched terminal **164***c* is connected to the first relay unit output **74**.

(28) The fourth relay **166** has a pole terminal **166***a*, a first switched terminal **166***b*, and a second switched terminal **166***c*. The pole terminal **166***a* is connected to the second terminal **162***b* of the fourth stator winding coil **34**. The first switched terminal **166***b* is connected to the first relay unit output **74**. The second switched terminal **166***c* is connected to the second terminal **168***b* of the fifth stator winding coil **36**.

(29) The fifth relay **170** has a pole terminal **170***a*, a first switched terminal **170***b*, and a second switched terminal **170***c*. The pole terminal **170***a* is connected to the first terminal **168***a* of the fifth stator winding coil **36**. The first switched terminal **170***b* is connected to first terminal **174***a* of the sixth stator winding coil **38**. The second switched terminal **170***c* is connected to the third relay unit output **78**.

(30) The sixth relay **172** has a first terminal **172***a* and a second terminal **172***b*. The first terminal **172***a* is connected to the second terminal **168***b* of the fifth stator winding coil **36**. The second terminal **172***b* is connected to the second relay unit output **76**.

(31) The seventh relay **176** has a first terminal **176***a* and a second terminal **176***b*. The first terminal **176***a* is connected to the first terminal **174***a* of the sixth stator winding coil **38**. The second terminal **176***b* is connected to the second relay unit output **76**.

(32) The eighth relay **178** has a pole terminal **178***a*, a first switched terminal **178***b*, and a second switched terminal **178***c*. The pole terminal **178***a* is connected to the second terminal **174***b* of the sixth stator winding coil **38**. The first switched terminal **178***b* is connected to the second relay unit output **76**. The second switched terminal **178***c* is connected to the second terminal **168***b* of the fifth

stator winding coil **36**.

(33) In the exemplary embodiment shown in FIG. **3**, the relay unit **16** is configured for 8-pole operation of the doubly-fed induction machine **12**. By switching each relay **156**, **160**, **164**, **166**, **170**, **172**, **176**, **178** of the relay unit **16**, the three-phase stator windings **24**, **26** may be connected for 4-pole operation of the doubly-fed induction machine **12**, as will be discussed in greater detail below.

(34) Referring to FIG. **4**, a block diagram of an alternate system for converting energy is illustrated and generally indicated by reference numeral **200**. The system **200** is similar to the system **10** shown in FIG. **1** and like components are indicated by like reference numbers. However, in the system **200**, the power conversion unit **14** of the system **10** is replaced by an alternate power conversion unit **202**.

(35) The alternate power conversion unit **202** is used to convert a direct current (DC) input power source to two three-phase alternating current (AC) output power sources and provide a three-phase AC power source to each of the three-phase rotor windings **40**, **42**. In the exemplary embodiment shown in FIG. **4**, the alternate power conversion unit **202** includes a stationary power transfer inverter **204**, a stationary power transfer coil **206**, a rotating power transfer coil **208**, a rotating power transfer rectifier **210**, a first inverter **68**, and a second inverter **70**. The alternate power conversion unit **202** also includes a stationary wireless data transceiver **212**, a rotating wireless data transceiver **214**, and a rotating controller **216**. The alternate power conversion unit **202** is in electrical communication with the controller **18** and the DC power supply **72***a*, **72***b*.

(36) The stationary power transfer inverter **204** is used to convert the DC power supply **72***a*, **72***b* to AC power for wireless power transfer using the stationary power transfer coil **206**. In an exemplary embodiment, the stationary power transfer inverter **204** is affixed to the stator **20** or an outer casing (not shown) of the doubly-fed induction machine **12** proximal to an end of the rotor **22**. The stationary power transfer inverter **204** is a square wave, modified sine wave, pulsed sine wave, pulse width modulated (PWM) wave or pure sine wave inverter. It should be understood that additional types of inverters or other electrical devices designed to convert DC power to AC power are within the scope of the present disclosure. The stationary power transfer inverter **204** is connected to the DC power supply **72***a*, **72***b* and the stationary power transfer coil **206**.

(37) The stationary power transfer coil **206** is used to wirelessly transfer power from the stationary power transfer inverter **204** to the rotating power transfer rectifier **210**. In an exemplary embodiment, the stationary power transfer coil **206** is one coil of a rotary transformer. The stationary power transfer coil **206** is affixed to the stator **20** or an outer casing (not shown) of the doubly-fed induction machine **12** proximal to an end of the rotor **22**. The stationary power transfer coil **206** is connected to the stationary power transfer inverter **204**, and inductively coupled with the rotating power transfer coil **208**. The stationary power transfer coil **206** is excited with a changing electric current by the stationary power transfer inverter **204**. The changing electric current induces a changing magnetic field near the stationary power transfer coil **206**.

(38) The rotating power transfer coil **208** is used to wirelessly receive power from the stationary power transfer inverter **204** via the stationary power transfer coil **206**. In an exemplary embodiment, the rotating power transfer coil **208** is one coil of a rotary transformer. The rotating power transfer coil **208** is affixed to the rotor **22** (e.g., the rotor shaft **144**) of the doubly-fed induction machine **12**. The rotating power transfer coil **208** is connected to the rotating power transfer rectifier **210**, and inductively coupled with the stationary power transfer coil **206**. The changing magnetic field induced by the stationary power transfer coil **206** induces an AC power in the rotating power transfer coil **208**. The AC power is converted to a DC power by the rotating power transfer rectifier **210**.

(39) The rotating power transfer rectifier **210** is used to convert the AC power induced in the rotating power transfer coil **208** to DC power. In an exemplary embodiment, the rotating power transfer rectifier **210** is affixed to the rotor **22** (e.g., the rotor shaft **144**) of the doubly-fed induction

machine **12**. The rotating power transfer rectifier **210** is a rectifier, for example, a controlled rectifier, an uncontrolled rectifier, a half-wave rectifier, a full-wave rectifier, and/or a bridge rectifier. It should be understood that additional types of rectifiers or other electrical devices designed to convert AC power to DC power are within the scope of the present disclosure. The rotating power transfer rectifier **210** is connected to the rotating power transfer coil **208** and the first and second inverters **68**, **70**.

(40) In an alternative exemplary embodiment, capacitive coupling may be used instead of inductive coupling to transfer power from the stationary power transfer inverter **204** to the rotating power transfer rectifier **210**. In the aforementioned alternative exemplary embodiment, a stationary conductive electrode (not shown) is affixed to the stator **20** or an outer casing (not shown) of the doubly-fed induction machine **12** proximal to an end of the rotor **22** and electrically connected to the stationary power transfer inverter **204**. A rotating conductive electrode (not shown) is affixed to the rotor **22** (e.g., the rotor shaft **144**) of the doubly-fed induction machine **12** and is electrically connected to the rotating power transfer rectifier **210**. The stationary conductive electrode and the rotating conductive electrode are capacitively coupled, allowing for wireless power transmission between the stationary power transfer inverter **204** and the rotating power transfer rectifier **210**.

(41) The inverters **68**, **70** convert DC power to AC power and allow control of various characteristics of the AC power by the controller **18**. In a non-limiting example, the inverters **68**, **70** are affixed to the rotor **22** (e.g., the rotor shaft **144**) of the doubly-fed induction machine **12**. The inverters **68**, **70** respond to data signals from the rotating controller **216** by adjusting an amplitude, frequency, and/or phase angle of the AC current and/or voltage output. The inverters **68**, **70** receive DC power from the rotating power transfer rectifier **210** and provide an AC power to each of the rotor winding terminals **56**, **58**, **60**, **62**, **64**, **66**. The inverters **68**, **70** are also in electrical communication with the rotating controller **216**. In an exemplary embodiment, the inverters **68**, **70** are square wave, modified sine wave, pulsed sine wave, pulse width modulated (PWM) wave or pure sine wave inverters. It should be understood that additional types of inverters or other electrical devices designed to convert DC power to AC power are within the scope of the present disclosure.

(42) The stationary wireless data transceiver **212** is used to transmit data from the controller **18** to the rotating controller **216**. In an exemplary embodiment, the stationary wireless data transceiver **212** is affixed to the stator **20** or an outer casing (not shown) of the doubly-fed induction machine **12** proximal to an end of the rotor **22**. In another exemplary embodiment, the stationary wireless data transceiver **212** is located within the controller **18**. The stationary wireless data transceiver **212** is a device capable of wirelessly transmitting and receiving data, for example, a WiFi transceiver, a Bluetooth transceiver, a radio-frequency transceiver, and/or the like. It should be understood that various additional types and protocols of wireless data transmission are within the scope of the present disclosure.

(43) The rotating wireless data transceiver **214** is used by the rotating controller **216** to receive data from the controller **18**. In an exemplary embodiment, the rotating wireless data transceiver **214** is affixed to the rotor **22** (e.g., the rotor shaft **144**) of the doubly-fed induction machine **12**. In another exemplary embodiment, the rotating wireless data transceiver **214** is located within the rotating controller **216**. The rotating wireless data transceiver **214** is a device capable of wirelessly transmitting and receiving data, for example, a WiFi transceiver, a Bluetooth transceiver, a radio-frequency transceiver, and/or the like. It should be understood that various additional types and protocols of wireless data transmission are within the scope of the present disclosure.

(44) The rotating controller **216** is used to process and take action based on data received by the rotating wireless data transceiver **214**. The rotating controller **216** is analogous to the controller **18**, and likewise contains a processor (not shown) and media (not shown) as discussed above in reference to the controller **18**. The rotating controller **216** is in electrical communication with the rotating wireless data transceiver **214**, the first inverter **68**, and the second inverter **70**. In a non-

limiting example, the rotating controller **216** may receive data from the controller **18** with instructions to adjust an amplitude, frequency, and/or phase angle of the AC current and/or voltage output of the first and second inverters **68**, **70**.

(45) In an exemplary embodiment, the system **10** and/or the system **200** is used as an aviation generator in an airliner. The rotor shaft **144** is mechanically affixed to a rotating member of a jet engine of the airliner, providing rotational energy to the rotor **22**. A DC power source of the airliner provides the DC power source **72***a*, **72***b* for the power conversion unit **14** and/or the alternate power conversion unit **202**. The relay unit outputs **74**, **76**, **78** are connected to a three-phase AC power system of the airliner having multiple electrical loads. It is advantageous for the three-phase AC power system to be supplied with power having a narrow frequency band. A rotational velocity of the jet engine and thus a rotational velocity of the rotor shaft **144** may vary during operation of the airliner, thus the operation of the doubly-fed induction machine **12** must be adjusted to regulate a voltage magnitude and frequency of the output power.

(46) To regulate the voltage magnitude of the output power, the controller **18** communicates with the inverters **68**, **70** to increase or decrease an excitation current magnitude of the three-phase rotor windings **40**, **42**, thereby regulating the voltage magnitude of the output power.

(47) To regulate the frequency of the output power, the media **82** of the controller **18** contains software instructions (i.e., the controller **18** is programmed) to determine a rotational velocity of the rotor shaft **144** and adjust the operation of the doubly-fed induction machine **12** based on the rotational velocity of the rotor shaft **144**. In a non-limiting example, the rotational velocity of the rotor shaft **144** is determined using a resolver, which is a rotary transformer used for measuring degrees of rotation. In another non-limiting example, the rotational velocity of the rotor shaft **144** is determined using a rotary or pulse encoder. It should be understood that various additional devices and methods which may be used by the controller **18** to determine the rotational velocity of the rotor shaft **144** are within the scope of the present disclosure.

(48) To adjust the operation of the doubly-fed induction machine **12** based on the rotational velocity of the rotor shaft **144**, the media **82** of the controller **18** contains software instructions (i.e., the controller **18** is programmed) to adjust an excitation frequency of the three-phase rotor windings **40**, **42** and/or adjust a pole number (i.e., a number of magnetic poles induced by electrical current flow through the three-phase rotor windings **40**, **42**) of the doubly-fed induction machine **12**. To adjust the excitation frequency of the three-phase rotor windings **40**, **42**, the controller **18** communicates with the inverters **68**, **70** of the power conversion unit **14** and/or the alternate power conversion unit **202** using a wired or wireless communication method as discussed above to command the inverters **68**, **70** to increase or decrease the excitation frequency of the three-phase rotor windings **40**, **42**. To adjust the pole number, the controller **18** communicates with the inverters **68**, **70** of the power conversion unit **14** and/or the alternate power conversion unit **202** using a wired or wireless communication method as discussed above to command the inverters **68**, **70** to adjust a plurality of phase angles of the excitation of the three-phase rotor windings **40**, **42**. The controller **18** additionally communicates with the relay unit **16** to connect the three-phase stator windings **24**, **26** for 8-pole or 4-pole operation of the doubly-fed induction machine **12**, as described above.

(49) Referring to FIG. **5**, a graph of a mode of operation of the system **10** and/or the system **200** according to an exemplary embodiment is illustrated and generally indicated by reference numeral **300**. The vertical axis **302** of the graph **300** indicates electrical frequency in units of hertz (Hz). The horizontal axis **304** of the graph **300** indicates the rotational velocity of the rotor shaft **144** in units of thousands of revolutions per minute (krpm). The first line **306** is the frequency of the output power as a function of the rotational velocity of the rotor shaft **144**. The second line **308** is the excitation frequency of the three-phase rotor windings **40**, **42** with the doubly-fed induction machine **12** operating in 8-pole mode. The third line **310** is the excitation frequency of the three-phase rotor windings **40**, **42** with the doubly-fed induction machine **12** operating in 4-pole mode.

For the second line **308** and the third line **310**, a negative excitation frequency means that the magnetic field generated by the three-phase rotor windings **40**, **42** is rotating in the opposite direction of the mechanical rotation of the rotor shaft **144**.
(50) In an exemplary embodiment, the excitation frequency of the three-phase rotor windings **40**, **42** and the pole number with which to operate the doubly-fed induction machine **12** are determined using the following formula:
(51) $f_{\text{statorwindings}} = f_{\text{rotorwindings}} + \frac{P * V_R}{120}$ wherein f.sub.stator windings is the frequency of the output power (i.e., first line **306**), f.sub.rotor windings is the excitation frequency of the three-phase rotor windings **40**, **42** (i.e., the second line **308** and the third line **310**), P is the pole number of the doubly-fed induction machine **12** (e.g., 4 or 8), and V.sub.R is the rotational velocity of the rotor shaft **144** in units of revolutions per minute (rpm).
(52) In the exemplary embodiment depicted by FIG. **5**, the pole number is switched from 8-pole to 4-pole when the rotor shaft **144** is rotating at **8** krpm. It should be understood that alternative embodiments may use other pole-switching combinations (e.g., 12-pole to 6-pole or 4-pole to 2-pole). To operate the doubly-fed induction machine **12** in 8-pole mode, the controller **18** commands the inverters **68**, **70** to excite the first rotor winding coil **44** and the fourth rotor winding coil **50** with a first sinusoidal current, the second rotor winding coil **46** and the fifth rotor winding coil **52** with a second sinusoidal current, and the third rotor winding coil **48** and the sixth rotor winding coil **54** with a third sinusoidal current. The second sinusoidal current lags the first sinusoidal current by 120 degrees. The third sinusoidal current lags the second sinusoidal current by 120 degrees. The controller **18** additionally commands the relay unit **16** to connect the three-phase stator windings **24**, **26** for 8-pole operation.
(53) To configure the relay unit **16** for 8-pole operation, the first relay **156** connects the pole terminal **156***a* to the first switched terminal **156***b*. The second relay **160** connects the pole terminal **160***a* to the first switched terminal **160***b*. The third relay **164** connects the pole terminal **164***a* to the first switched terminal **164***b*. The fourth relay **166** connects the pole terminal **166***a* to the first switched terminal **166***b*. The fifth relay **170** connects the pole terminal **170***a* to the first switched terminal **170***b*. The sixth relay **172** connects the first terminal **172***a* to the second terminal **172***b*. The seventh relay **176** disconnects the first terminal **176***a* from the second terminal **176***b*. The eighth relay **178** connects the pole terminal **178***a* to the first switched terminal **178***b*.
(54) To operate the doubly-fed induction machine **12** in 4-pole mode, the controller **18** commands the first inverter **68** to excite the first rotor winding coil **44** with a fourth sinusoidal current, the second rotor winding coil **46** with a fifth sinusoidal current, and the third rotor winding coil **48** with a sixth sinusoidal current. The controller **18** additionally commands the second inverter **70** to excite the fourth rotor winding coil **50** with a seventh sinusoidal current, the fifth rotor winding coil **52** with an eighth sinusoidal current, and the sixth rotor winding coil **54** with a ninth sinusoidal current. The fifth sinusoidal current lags the fourth sinusoidal current by 240 degrees. The sixth sinusoidal current lags the fourth sinusoidal current by 120 degrees. The seventh sinusoidal current is out of phase with the fourth sinusoidal current by 180 degrees. The eighth sinusoidal current is out of phase with the fifth sinusoidal current by 180 degrees. The ninth sinusoidal current is out of phase with the sixth sinusoidal current by 180 degrees. The controller **18** additionally commands the relay unit **16** to connect the three-phase stator windings **24**, **26** for 4-pole operation.
(55) To configure the relay unit **16** for 4-pole operation, the first relay **156** connects the pole terminal **156***a* to the second switched terminal **156***c*. The second relay **160** connects the pole terminal **160***a* to the second switched terminal **160***c*. The third relay **164** connects the pole terminal **164***a* to the second switched terminal **164***c*. The fourth relay **166** connects the pole terminal **166***a* to the second switched terminal **166***c*. The fifth relay **170** connects the pole terminal **170***a* to the second switched terminal **170***c*. The sixth relay **172** disconnects the first terminal **172***a* from the second terminal **172***b*. The seventh relay **176** connects the first terminal **176***a* to the second terminal **176***b*. The eighth relay **178** connects the pole terminal **178***a* to the second switched

terminal **178***c*.

(56) In an additional exemplary embodiment, the doubly-fed induction machine **12** is used as a motor to provide rotational energy to the jet engine of the airliner during a startup process of the jet engine. In additional exemplary embodiments, the system **10** or the system **200** is used in a vehicle, such as an automobile, and the rotor shaft **144** is connected to a drivetrain of the automobile (e.g., using a transmission, a gearbox, and/or other means of transferring mechanical power). It should be understood that additional uses of the system **10**, **200** and/or the doubly-fed induction machine **12** to convert energy are included in the scope of the present disclosure.

(57) The system **10**, **200** of the present disclosure offers several advantages. When used as a generator, the frequency of the output power can be tightly regulated. In the exemplary embodiment depicted in FIG. **5**, the output frequency remains between 360 Hz and 440 Hz over a large range of velocities of the rotor shaft **144**. Additionally, because the excitation frequency of the three-phase rotor windings **40**, **42** is much lower than the frequency of the power induced on the three-phase stator windings **24**, **26**, the inverters **68**, **70** need only be rated for a fraction of the power output of the systems **10**, **200** as given by the following formula:

(58) $\frac{P_{statorwindings}}{P_{rotorwindings}} = \frac{f_{statorwindings}}{f_{rotorwindings}}$ where P.sub.stator windings is the power generated at the three-phase stator windings **24**, **26**, P.sub.rotor windings is the power supplied to the three-phase rotor windings **40**, **42**, f.sub.stator windings is the frequency of the output power, and f.sub.rotor windings is the excitation frequency of the rotor windings **40**, **42**.

(59) Additionally, the systems **10**, **200** may be immediately de-energized in an emergency situation by disabling the inverters **68**, **70**. Disabling the inverters **68**, **70** will stop the generation of power even if the rotor **22** is still rotating.

(60) The description of the present disclosure is merely exemplary in nature and variations that do not depart from the gist of the present disclosure are intended to be within the scope of the present disclosure. Such variations are not to be regarded as a departure from the spirit and scope of the present disclosure.

Claims

1. A system for converting energy, the system comprising: a doubly-fed induction machine including: a rotor having a plurality of rotor windings; and a stator having a plurality of stator windings; a power conversion unit in electrical communication with the plurality of rotor windings, wherein the power conversion unit is configured to excite the plurality of rotor windings with an alternating current; a relay unit in electrical communication with the plurality of stator windings, wherein the relay unit is configured to electrically interconnect a plurality of terminals of the plurality of stator windings; and a controller in electrical communication with the power conversion unit and the relay unit, wherein the controller is programmed to: determine a rotational velocity of the rotor; and adjust an operation of the doubly-fed induction machine based on the rotational velocity of the rotor by using the power conversion unit to excite the plurality of rotor windings with one of: a first phase configuration to provide a first quantity of effective poles and a second phase configuration to provide a second quantity of effective poles and by using the relay unit to electrically interconnect one or more of the plurality of terminals of the plurality of stator windings in one of: a first connection configuration to provide the first quantity of effective poles and a second connection configuration to provide the second quantity of effective poles.

2. The system of claim 1, wherein: the plurality of rotor windings further includes: a first three-phase rotor winding having a first rotor winding coil, a second rotor winding coil, and a third rotor winding coil; and a second three-phase rotor winding having a fourth rotor winding coil, a fifth rotor winding coil, and a sixth rotor winding coil; the plurality of stator windings further includes: a first three-phase stator winding having a first stator winding coil having a first and second terminal,

a second stator winding coil having a first and second terminal, and a third stator winding coil having a first and second terminal; and a second three-phase stator winding having a fourth stator winding coil having a first and second terminal, a fifth stator winding coil having a first and second terminal, and a sixth stator winding coil having a first and second terminal; and the system further includes: a first, second, and third output terminal.

3. The system of claim 2, wherein the power conversion unit further comprises: a first inverter having three outputs, wherein each output of the first inverter is in electrical communication with one of the first, second, or third rotor winding coils; a second inverter having three outputs, wherein each output of the second inverter is in electrical communication with one of the fourth, fifth, or sixth rotor winding coils; and wherein the first inverter and the second inverter are configured to be powered by a direct current (DC) power source.

4. The system of claim 3 further comprising: a slip ring configured to provide electrical communication between each output of the first inverter and one of the first, second, or third rotor winding coils and between each output of the second inverter and one of the fourth, fifth, or sixth rotor winding coils.

5. The system of claim 3, wherein the relay unit further comprises: a plurality of solid-state relays in electrical communication with the first three-phase stator winding and the second three-phase stator winding.

6. The system of claim 3, wherein to adjust the operation of the doubly-fed induction machine, the controller is further programmed to: adjust an excitation frequency of the first three-phase rotor winding using the first inverter and the excitation frequency of the second three-phase rotor winding using the second inverter of the power conversion unit based on the rotational velocity of the rotor to regulate an output voltage frequency when the doubly-fed induction machine is used as a generator.

7. The system of claim 3, wherein to adjust the operation of the doubly-fed induction machine, the controller is further programmed to: configure the doubly-fed induction machine to have four effective poles using the power conversion unit and the relay unit based on the rotational velocity of the rotor to regulate an output voltage frequency when the doubly-fed induction machine is used as a generator; and configure the doubly-fed induction machine to have eight effective poles using the power conversion unit and the relay unit based on the rotational velocity of the rotor to regulate an output voltage frequency when the doubly-fed induction machine is used as a generator.

8. The system of claim 7, wherein to configure the doubly-fed induction machine to have four effective poles, the controller is further programmed to: configure the first inverter to: excite the first rotor winding coil with a first sinusoidal current; excite the second rotor winding coil with a second sinusoidal current, wherein the second sinusoidal current lags the first sinusoidal current by 240 degrees; excite the third rotor winding coil with a third sinusoidal current, wherein the third sinusoidal current lags the first sinusoidal current by 120 degrees; configure the second inverter to: excite the fourth rotor winding coil with a fourth sinusoidal current, wherein the fourth sinusoidal current is 180 degrees out of phase with the first sinusoidal current; excite the fifth rotor winding coil with a fifth sinusoidal current, wherein the fifth sinusoidal current is 180 degrees out of phase with the second sinusoidal current; excite the sixth rotor winding coil with a sixth sinusoidal current, wherein the sixth sinusoidal current is 180 degrees out of phase with the third sinusoidal current; and configure the relay unit to: electrically connect the first terminal of the first stator winding coil with the second output terminal and the first terminal of the sixth stator winding coil; electrically connect the first terminal of the second stator winding coil with the third output terminal and the first terminal of the fifth stator winding coil; electrically connect the first terminal of the third stator winding coil with the first output terminal and the first terminal of the fourth stator winding coil; and electrically connect the second terminal of the fourth stator winding coil to the second terminal of the fifth stator winding coil and the second terminal of the sixth stator winding coil.

9. The system of claim 7, wherein configure the doubly-fed induction machine to have eight effective poles, the controller is further programmed to: configure the first inverter to: excite the first rotor winding coil with a first sinusoidal current; excite the second rotor winding coil with a second sinusoidal current, wherein the second sinusoidal current lags the first sinusoidal current by 120 degrees; excite the third rotor winding coil with a third sinusoidal current, wherein the third sinusoidal current lags the second sinusoidal current by 120 degrees; configure the second inverter to: excite the fourth rotor winding coil with the first sinusoidal current; excite the fifth rotor winding coil with the second sinusoidal current; excite the sixth rotor winding coil with the third sinusoidal current; and configure the relay unit to: electrically connect the first terminal of the first stator winding coil with the third output terminal and the second terminal of the sixth stator winding coil; electrically connect the first terminal of the second stator winding coil with the second output terminal and the second terminal of the fifth stator winding coil electrically connect the first terminal of the third stator winding coil with the first output terminal and the second terminal of the fourth stator winding coil; and electrically connect the first terminal of the fourth stator winding coil to the first terminal of the fifth stator winding coil and the first terminal of the sixth stator winding coil.
10. A system for converting energy for a vehicle, the system comprising: a doubly-fed induction machine including: a rotor having a plurality of rotor windings, wherein the plurality of rotor windings further includes a first three-phase rotor winding having a first rotor winding coil, a second rotor winding coil, and a third rotor winding coil and a second three-phase rotor winding having a fourth rotor winding coil, a fifth rotor winding coil, and a sixth rotor winding coil; and a stator having a plurality of stator windings; a power conversion unit in electrical communication with the plurality of rotor windings, the power conversion unit further comprising: a first inverter having three outputs, wherein each output of the first inverter is in electrical communication with one of the first, second, or third rotor winding coils; and a second inverter having three outputs, wherein each output of the second inverter is in electrical communication with one of the fourth, fifth, or sixth rotor winding coils; a relay unit in electrical communication with the plurality of stator windings; and a controller in electrical communication with the power conversion unit and the relay unit, wherein the controller is programmed to: determine a rotational velocity of the rotor; adjust an operation of the doubly-fed induction machine using the power conversion unit and the relay unit based on the rotational velocity of the rotor, wherein to adjust the operation of the doubly-fed induction machine, the controller is further programmed to: configure the doubly-fed induction machine to have four effective poles using the power conversion unit by: configuring the first inverter to excite the first rotor winding coil with a first sinusoidal current, excite the second rotor winding coil with a second sinusoidal current, wherein the second sinusoidal current lags the first sinusoidal current by 240 degrees, excite the third rotor winding coil with a third sinusoidal current, wherein the third sinusoidal current lags the first sinusoidal current by 120 degrees; and configuring the second inverter to excite the fourth rotor winding coil with a fourth sinusoidal current, wherein the fourth sinusoidal current is 180 degrees out of phase with the first sinusoidal current, excite the fifth rotor winding coil with a fifth sinusoidal current, wherein the fifth sinusoidal current is 180 degrees out of phase with the second sinusoidal current, and excite the sixth rotor winding coil with a sixth sinusoidal current, wherein the sixth sinusoidal current is 180 degrees out of phase with the third sinusoidal current.
11. The system of claim 10, wherein: the plurality of stator windings further includes: a first three-phase stator winding having a first stator winding coil having a first and second terminal, a second stator winding coil having a first and second terminal, and a third stator winding coil having a first and second terminal; and a second three-phase stator winding having a fourth stator winding coil having a first and second terminal, a fifth stator winding coil having a first and second terminal, and a sixth stator winding coil having a first and second terminal; and the system further includes: a first, second, and third output terminal.

12. The system of claim 11 wherein the first inverter and the second inverter are affixed to the rotor, and wherein the power conversion unit further comprises: a wireless power transfer system including: a stationary power transfer inverter; a stationary wireless power transfer coil in electrical communication with the stationary power transfer inverter; a rotating power transfer rectifier affixed to the rotor; a rotating wireless power transfer coil affixed to the rotor, wherein the rotating wireless power transfer coil is in electrical communication with the rotating power transfer rectifier and the first inverter and the second inverter of the power conversion unit; and wherein the stationary wireless power transfer coil is inductively coupled to the rotating wireless power transfer coil to transmit power between the stationary power transfer inverter and the rotating power transfer rectifier; and a wireless data transfer system including: a stationary wireless data transceiver; a rotating wireless data transceiver affixed to the rotor; and a rotating controller affixed to the rotor, wherein the rotating controller is in electrical communication with the rotating wireless data transceiver and the first inverter and the second inverter of the power conversion unit, and wherein the rotating controller is programmed to: receive command data from the stationary wireless data transceiver using the rotating wireless data transceiver; and adjust an operation of the first inverter and the second inverter of the power conversion unit based at least in part on the command data.
13. The system of claim 11, wherein the relay unit further comprises: a plurality of solid-state relays in electrical communication with the first three-phase stator winding and the second three-phase stator winding.
14. The system of claim 11, wherein to adjust the operation of the doubly-fed induction machine, the controller is further programmed to: adjust an excitation frequency of the first three-phase rotor winding using the first inverter and the excitation frequency of the second three-phase rotor winding using the second inverter of the power conversion unit based on the rotational velocity of the rotor to regulate an output voltage frequency when the doubly-fed induction machine is used as a generator.
15. The system of claim 11, wherein to adjust the operation of the doubly-fed induction machine, the controller is further programmed to: configure the doubly-fed induction machine to have four effective poles using the power conversion unit and the relay unit based on the rotational velocity of the rotor to regulate an output voltage frequency when the doubly-fed induction machine is used as a generator; and configure the doubly-fed induction machine to have eight effective poles using the power conversion unit and the relay unit based on the rotational velocity of the rotor to regulate an output voltage frequency when the doubly-fed induction machine is used as a generator.
16. The system of claim 15, wherein configure the doubly-fed induction machine to have four effective poles, the controller is further programmed to: configure the relay unit to: electrically connect the first terminal of the first stator winding coil with the second output terminal and the first terminal of the sixth stator winding coil; electrically connect the first terminal of the second stator winding coil with the third output terminal and the first terminal of the fifth stator winding coil electrically connect the first terminal of the third stator winding coil with the first output terminal and the first terminal of the fourth stator winding coil; and electrically connect the second terminal of the fourth stator winding coil to the second terminal of the fifth stator winding coil and the second terminal of the sixth stator winding coil.
17. The system of claim 15, wherein configure the doubly-fed induction machine to have eight effective poles, the controller is further programmed to: configure the first inverter to: excite the first rotor winding coil with a first sinusoidal current; excite the second rotor winding coil with a second sinusoidal current, wherein the second sinusoidal current lags the first sinusoidal current by 120 degrees; excite the third rotor winding coil with a third sinusoidal current, wherein the third sinusoidal current lags the second sinusoidal current by 120 degrees; configure the second inverter to: excite the fourth rotor winding coil with the first sinusoidal current; excite the fifth rotor winding coil with the second sinusoidal current; excite the sixth rotor winding coil with the third

sinusoidal current; and configure the relay unit to: electrically connect the first terminal of the first stator winding coil with the third output terminal and the second terminal of the sixth stator winding coil; electrically connect the first terminal of the second stator winding coil with the second output terminal and the second terminal of the fifth stator winding coil electrically connect the first terminal of the third stator winding coil with the first output terminal and the second terminal of the fourth stator winding coil; and electrically connect the first terminal of the fourth stator winding coil to the first terminal of the fifth stator winding coil and the first terminal of the sixth stator winding coil.

18. An electric machine comprising: a stator having a hollow cylindrical shape, the stator including: a plurality of stator slots disposed at a uniform pitch along a circumference of the stator; a first, second, third, fourth, fifth, and sixth stator coil, wherein each stator coil has a first stator coil loop electrically connected in series to a second stator coil loop; and wherein for each of the first, second, third, fourth, fifth, and sixth stator coil: the first stator coil loop is disposed in a first pair of stator slots, wherein a center of the first stator coil loop is substantially coincident with the circumference of the stator, and wherein a second pair of stator slots is disposed between the first pair of stator slots; and the second stator coil loop is disposed in a third pair of stator slots, wherein a center of the second stator coil loop is substantially coincident with the circumference of the stator, wherein a fourth pair of stator slots is disposed between the third pair of stator slots, and wherein the third pair of stator slots are diametrically opposed to the first pair of stator slots such that the second stator coil loop is diametrically opposed to the first stator coil loop; and a rotor disposed rotatably inside the stator along a central axis of the stator, the rotor including: a plurality of rotor slots disposed at a uniform pitch along a circumference of the rotor; a first, second, third, fourth, fifth, and sixth rotor coil, wherein each rotor coil has a first rotor coil loop electrically connected in series to a second rotor coil loop; and wherein for each of the first, second, third, fourth, fifth, and sixth rotor coil: the first rotor coil loop is disposed in a first pair of rotor slots, wherein a center of the first rotor coil loop is substantially coincident with the circumference of the rotor, and wherein a second pair of rotor slots is disposed between the first pair of rotor slots; and the second rotor coil loop is disposed in a third pair of rotor slots, wherein a center of the second rotor coil loop is substantially coincident with the circumference of the rotor, wherein a fourth pair of rotor slots is disposed between the third pair of rotor slots, and wherein the third pair of rotor slots are diametrically opposed to the first pair of rotor slots such that the second rotor coil loop is diametrically opposed to the first rotor coil loop.

19. The electric machine of claim 18, further including a controller affixed to an exterior surface of the stator, wherein the controller is programmed to: determine a rotational velocity of the rotor; and electrically change a pole number of the rotor and the stator based at least in part on the rotational velocity of the rotor by adjusting a phase angle of each phase of a three-phase current supply electrically connected to the rotor coils and changing a plurality of electrical connections between each of the stator coils.